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CART 263-B

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Utilizing Code to Honour an Artist's Legacy

The early stage of brainstorming for our project was quite challenging. The assignment gave us a great deal of freedom and did not impose any limitations on the theme, which unexpectedly made the process more difficult. Without clear constraints, it was hard for us to come up with a solid and unique idea.

At that point, Pansy suggested that we should choose a topic that we both genuinely enjoy. By doing so, we would naturally feel more passionate and motivated about working on the project. After discussing it for a while, she thought of a music artist that both of us deeply admire, who unfortunately passed away several years ago. We quickly agreed that creating something artistic to commemorate him would be meaningful.

The artist is Wowaka, a Japanese producer and musician well known for his Vocaloid songs. Vocaloid is a voice synthesizer software that generates artificial singing voices. Because Vocaloid itself is a digital program, it also felt conceptually fitting for our programming-based project. [1]

Wowaka's album *Unhappy Refrain* became a major source of inspiration for us.[2] Each song in the album has its own unique cover artwork, yet they all share a consistent visual

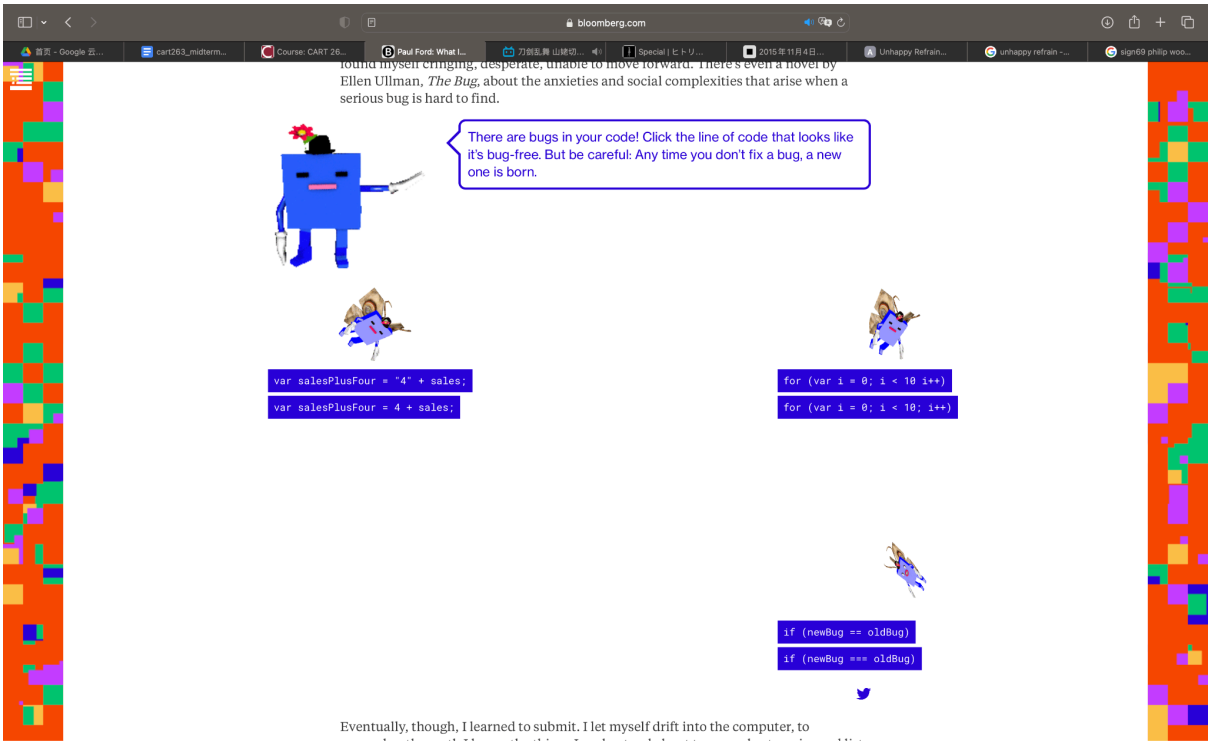
[1] Hitorie Atelier.
[2] Wikipedia, "Unhappy Refrain."

style. The covers are composed of simple geometric shapes and low-saturation colors. The minimalist design creates a strong visual identity while remaining visually striking. From a technical perspective, we also felt that patterns built from simple geometric shapes would be relatively feasible to recreate through programming.

Our initial idea was therefore to create something animated and music-related, inspired by Wowaka's visual style. The plan was, each of us would select a song from him, and create an artistic audio-visual cue through coding.

While we were quickly browsing through the course readings, a section from Bloomberg.com caught our attention.[3] As shown in the screenshot, the butterfly animation on the page was very lively and visually engaging, yet it was created using relatively simple techniques. After analyzing it, I believe the animation was constructed using a sequence of PNG images combined with CSS-based distortion effects. The original PNG images appear to be stretched and rotated at different angles, creating the illusion that the butterfly is fluttering and flying. Despite its simplicity, the effect feels dynamic and expressive, which made it a useful reference for thinking about how we might implement animation in our own project.

[3] Paul Ford, "What is Code?"



We also gained inspiration from this webwork name *in:verse* made by Sukanya Aneja (artbase.rhizome.org). While exploring the archive, we found a web-based artwork that generates different visuals based on poems. The website creates a table of words consisting of poetic vocabulary, and each word is assigned to a mathematical or programming variable such as *sin*, *cos*, *min*, or *max*. By combining different words together, the system calculates different visual outputs.

What we found particularly interesting about this project is the way it translates language into visual forms. The poetic words themselves do not directly describe images, but through the assigned variables and calculations they influence how the graphics are generated. Different word combinations therefore produce different visual results.

This approach was very inspiring for our project because it demonstrates how one form of art can be transformed or manipulated through another medium. In our case, we were

also working with a similar idea: allowing music, another artistic form, to control and transform visual elements through code.

From there, we started to develop the idea of recreating the song cover using simple geometric shapes such as circles and rectangles, and then animating them according to the rhythm of the music. By doing this, we hoped to transform the originally static album artwork into something more dynamic and expressive. The movement of the shapes responding to the beat would allow the visual elements to interact directly with the music, creating a deeper connection between sound and image.

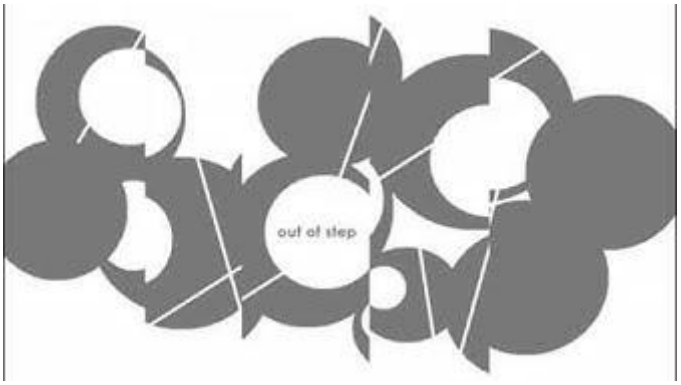
Another reason this idea felt meaningful to me was related to the artist himself. Since Wowaka has already passed away, many of his works now exist only as recordings and static images. By allowing the static cover artwork to move along with the rhythm of the music, it almost feels as if the music is coming to life again. The visuals are no longer just a background or decoration, but instead become something that reacts and breathes with the sound. In a way, it felt like a small artistic gesture to honor the artist and keep his work alive through motion and interaction.

We also came to the decision to split up the work between ourselves, as we thought that just one song would not be enough. Therefore, we each chose a song we felt passionate about and wanted to bring to life through programming. This next segment will split into our own point of views as we developed our ideas for each song.

Pansy: Zureteiku

While thinking about our concept, I also began to consider which specific song I wanted to work with. I eventually chose a song called “Zureteiku.” The title roughly

translates to “gradually out of step” . I found the title and the feeling of the song very interesting, because the idea of things slowly shifting out of alignment could also be expressed visually through animation. Shapes could subtly drift, shift, or move slightly off rhythm, reflecting the theme suggested by the song’s title. This gave me some early ideas about how the motion of the shapes might evolve throughout the piece.



My process began with trying to replicate the shapes of the cover image through code. At the beginning, both of us agreed that we should attempt to reproduce the entire image using JavaScript by breaking it down into many small circles. The idea was making each circle represent a small piece of the overall composition.

However, this approach quickly became quite difficult to manage. The original cover image contains a large number of semicircles and arcs, and recreating each of these shapes individually through code turned out to be much more complicated than I initially expected. If every shape needed to be decomposed into smaller circular elements, the number of layers increased dramatically. As the number of elements grew, the program became harder to control. Many of the layers began colliding with each other or appearing slightly mispositioned. Small calculation errors in coordinates would cause the shapes to overlap incorrectly, and adjusting each element manually was extremely time-consuming. I can also foresee how much trouble they will give me once I start to code the animation section. At

this stage, I realized that while the idea of fully reconstructing the image through JavaScript was conceptually interesting, it was not the most practical method for achieving the visual result we wanted.

This difficulty forced me to reconsider my approach and think about alternative ways to recreate the visual style while keeping the code manageable. Instead of trying to generate every shape entirely through JavaScript, I decided to simplify the structure by dividing the original cover image into five main transparent PNG layers. By doing this, I only needed to recreate three circles directly through code, which greatly reduced the overall complexity of the program.

After carefully analyzing the original artwork, I categorized the shapes into five visual groups. Each group was exported as a separate transparent PNG and moved as a single unit. The movement of these elements is also synchronized with the beat of the music, allowing the visuals to respond dynamically to the audio.

The part of the code that gathers data from the music and connects it to the visual animation was based on a music demo provided by Professor Sabine. Using this demo as a reference, we were able to extract audio data and use it to influence the animation parameters. The movement of the shapes becomes faster and more intense when the music itself becomes faster. This helped create a stronger connection between the sound and the visual motion, making the animation feel more responsive and alive.

Ben: Unknown Mother-Goose

Initially, I was thinking of choosing one of the songs that were covered within Wowaka's *Unhappy Refrain* album. However, I later decided against this and chose to tackle his very last song: Unknown Mother-Goose (or UMG for short).

UMG was his last song that was released back in August, 2017 as a part of Hatsune Miku's 10th-anniversary compilation album: *Re:Start*. The song marked his return to Vocaloid after a six-year long hiatus before his tragic passing, and is a tribute to his 'mother-figure' Hatsune Miku. It was because of her creation and popularity that allowed him to skyrocket as a producer/musician, and he owes his career to her success on a global scale that led to him being a prominent figure within the Vocaloid community. Just like her, he inspired millions of people to pursue their passion.

While I only discovered him recently thanks to Pansy, I have been a large fan of Miku and Vocaloid in general since 2011, which has gone on to influence a lot of what I do to this day. Despite being a recent fan of his, I can see how he influenced and changed the lives of so many, as his music hit me to the core. Therefore, I felt that it was fitting to cover this song through an artistic endeavor as tribute to the man who went on to inspire millions, despite being released many years after *Unhappy Refrain*.

It was through this thought process that I decided to approach the visuals for the song through surrounding real-world context, rather than digging deep into the lyrics and message of the song itself. The cover art for UMG depicts a fractured, abstract and geometrical heart with desaturated and shifting value hues that I used as my original basis for inspiration.

Afterwards, I did some digging through the different readings and one site in particular caught my eye: the Experiments with Google page. A page containing a collection

of different procedurally generated web art pages that change and shift through different conditions. Specifically, Seyhun Av Kim's WebGL Particle Audio Visualizer was a huge source of inspiration for what we decided to do[4]. What it does is pick up audio through your microphone and that audio then influences particle generation in a visually stunning way.

As stated prior, we decided to utilize Wowaka's existing songs to create and influence our audio visual creations, and I thought about going through a similar route to Kim's page for my song choice. However, even with help from Sabine, I did not have anywhere near the same level of experience or knowledge to create something even close to that level, so I decided to reconsider and take a different approach.

I rethought about the heart visual on UMG's cover art, and how it was Wowaka's last song before he passed. I also sought out some audio-visual inspiration on Pinterest and realized that many of the audio spectrum bars we see in audio-visual art resemble that of a pulse being displayed on a heart monitor, and I began to realize what I wanted to do. I utilized a heart graphic through an SVG to pump with the rhythm of the song, as well as an audio waveform on top. This was symbolic in a sense, as the heart and waveform represented the love for music he had and in turn how many people loved him through the music he created. However, this also represented a heart's pulse, that when the song ended, so too would the pulse dim and flatline, symbolizing his passing after the release of this song.

I fell in love with this idea and spent many hours on it, yet it proved to be either visually uninteresting or challenging. Unable to get the waveform to move in the way I wanted or just finding the visuals to be way too simple and not what I had imagined. I was already struggling as is: utilizing the analyzer node to influence the way in which the visual

[4] Chrome, "Experiments with Google."

elements shifted to the audio was difficult. I faltered and decided to once again take a different approach.

Taking a page out of Pansy's book, I decided to utilize UMG's cover art and turn it into an SVG so that I could manipulate it through Javascript. I added the main fragmented heart SVG into my DOM and was able to scale the different parts to the audio, as they shifted depending on the current frequency. This was also quite frustrating as I had a lot of issues adding and manipulating the SVG itself as it was injected directly into my html and not through a separate file. This turned out to be a bit simple but more visually appealing than what I had previously, and fit into the overall aesthetic of the project.

All in all, we mainly took aesthetic and functional inspiration from the readings rather than theoretical, in which we applied what we have found and learned to create a tribute to an artist we both love who met an unfortunate end all those years ago.

Main Sources

"Chrome Experiments - Experiments with Google." n.d. Experiments.withgoogle.com. https://experiments.withgoogle.com/collection/chrome.

"What is Code?" 2026. Bloomberg.com. 2026. https://www.bloomberg.com/graphics/2015-paul-ford-what-is-code/?embedded-checkout=true.

Wikipedia Contributors. 2026. "Unhappy Refrain." Wikipedia. Wikimedia Foundation. January 8, 2026.

"2015年11月4日発売 ヒトリエ「シャッタードール」特設サイト." 2026. 2015年11月4日発売 ヒトリエ「シャッタードール」特設サイト. 2026. https://www.hitorie.jp/static/special/shutter_doll/index.html.

Project URL: https://github.com/pansyqwq/Cart263_Project/tree/main/263-Midterm

Other External Links we Used

Wowaka Songs Playlist on Youtube:

https://www.youtube.com/playlist?list=PLhhV0tm_8OFe7UX7IF5EW_uXvzGdeYrPV

Rhizome Net Art in Readings for Minor Inspiration:

https://artbase.rhizome.org/wiki/Q12167